A prototype retrieval-based question and answering system based on a corpus of publicly available regulatory documents

PGDip Applied Data Science

PROJC9013 Applied Data Analytics Project

Interim Report

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Abstract

This project interim report explores the concept of a machine learning-based solution to enable a person to ask a question, and receive a natural language response based on the context of the question, where the answer to the question is expected to be derived from a specific knowledge-base of regulatory banking documents. It covers Literature Review, Core Architecture Design, Document Sourcing, Ingestion Pipeline and Initial Exploratory Data Analysis, with a Plan to complete the project.

1 Introduction

1.1 Background

This project builds on prior work completed by the author within this PGDip Applied Data Science programme. A preliminary study (SWRDC9011 Machine Learning and AI with Industrial Application CA1, November 2025) defined the organisational problem of context-sensitive retrieval of guidance from document repositories, proposed an end-to-end workflow for retrieval and generation, and identified key ethical and governance risks. A subsequent study (SWRDC9011 Machine Learning and AI with Industrial Application CA2, January 2026) validated PDF file extraction and performed exploratory data analysis on a small corpus enabling a metadata of document attributes to be assembled, and an assessment of document similarity.

1.2 Project Aim

Develop a prototype retrieval-based question and answering system based on a corpus of publicly available regulatory documents associated with the Irish banking industry. Given a natural-language user question, the system retrieves the most relevant document segments and generates an answer grounded exclusively in those segments, presenting citations (which document & document section) and, where applicable, highlighting conflicting guidance across retrieved sources. The system supports incremental ingestion of new documents with automated extraction, cleaning, metadata enrichment, chunking, and indexing. The working prototype could be a python-based web interface demonstrating system functionality. The intention is that the working prototype will be python-based web interface demonstrating end-to-end system functionality.

1.3 Primary Research Question

How effectively can a retrieval-based question and answering system grounded in banking-related regulatory documents support users in understanding Irish retail banking regulatory obligations?

Additionally, in answering this question, the project sets out to answer:

1. If new documents can be incorporated through an automated ingestion pipeline?
2. Is it possible to get accurate and reliable grounded answers with verifiable citations?
3. Can conflicting regulatory guidance be detected automatically and exposed to the user?

1.4 Hypotheses

Null Hypothesis (H0) A retrieval-augmented question and answer system does not produce more accurate, traceable, or reliable responses to regulatory queries than a standalone large language model.

Alternative Hypothesis (H1) A retrieval-augmented question and answer system produces significantly more accurate, traceable and reliable responses to regulatory queries than a standalone large language model.

This implies, for H0, that there is no statistically significant difference between a standalone LLM and a Retrieval Augmented Generation (RAG) system with respect to:

- answer correctness
- citation grounding
- hallucination rate
- usefulness to users

It follows, for H1, that the system shows statistically significant improvement in at least one of the above metrics.

1.5 Project Scope

The project will not build a fully scalable production-ready system. The system will be a prototype that is sufficient to meet the project aims and answer the defined research questions.

The project corpus for the project will be limited to fewer than 100 public regulatory documents applicable to banking in the Republic of Ireland, a scale that is considered sufficient to demonstrate core design, accuracy, and governance principles, with consideration for production scaling.

1.6 Interim Report Scope

This interim report is important in the delivery of the overall project as it enables

- validation of the project scope and hypothesis based on research and literature review
- data collection and exploratory data analysis
- evolution of design considerations based on work progressed to date
- a clear plan to complete the final project deliverable
- guidance and feedback to be taken on board from project lecturer/superviser to improve the final project deliverable

2 Research and Literature Review

It is important that the research and literature review is of direct relevance to the project scope, and of value to the design and implementation of the final system. On that basis, the research and literature review was conducted thematically, as follows:

- Regulatory Interpretation and need for traceable answers
- Sources and characteristics of regulatory documents
- Automated acquisition and preprocessing of regulatory documents
- Existing commercial solutions that support regulatory compliance
- AI approaches suitable for regulatory information retrieval

2.1 Regulatory Interpretation and need for traceable answers

There has been significant globalisation of 'rules' over the last 50 years, in particular concerning business regulation (1). But regulation in banking has existed for many years. European banking can be traced back to the third and second centuries BC to the island of Delos which became a centre of international banking for mediterranean trade. (1). National banks evolved into Central Banks and the globalisation of financial markets and their regulation.

Banks have an obligation to comply with regulations, which are laws determining how a bank must operate, ensuring the stability of financial institutions, and considering the interests of banking customers. Risk management is of fundamental importance in banking operations and compliance with regulations (2). Indeed, *Regulatory Risk* specifically looks at the likelihood of changes to banking operations and compliance obligations.

Following the Global Financial Crisis 2008 oversight and supervision of banks was significantly enhanced, ensuring that they comply with regulatory obligations and demonstrate prudential risk management (2).

As was noted in a previous paper by the author "The rules by which Irish banks are governed are set by a combination of EU legislation and national laws.(3). As Ireland is an EU Member State, the competent authority for banking regulation is the European Central Bank, which directly supervises companies designated as 'significant institutions', with the Central Bank of Ireland supervising 'less significant institutions', with regulation being mainly defined by the Single Supervisory Mechanism Mechanism (SSM) Framework Regulation and the Central Bank Acts. Supervision controls that a bank complies with its legal and regulatory obligations, with the sanctions for the company, and its personnel, for non-compliance."

It is essential for banks to implement an effective and robust risk management framework that includes assurance of an understanding of its regulatory obligations, and how these are transcribed into policies and operational procedures.

This is not straightforward. Identifying all relevant obligations is a challenge due to the complexity and volume of regulations (4). This typically requires a significant financial investment in compliance infrastructure. Additionally, regulations evolve and the implications of emerging obligations must be interpreted and impacts integrated to operations in line with compliance timescales. Advanced analytics, machine learning, and artificial intelligence can be exploited to improve the compliance environment.

A study in the language of written banking regulations (5) noted that the language is complex, typically because the subject matter is complex - it is describing the governance of a complex financial system. Many factors and eventualities have to be taken into account which lead to complex documents. Complex regulation requires more effort and information to understand, and there is a greater risk that it has not been interpreted correctly. Regulatory documents are often *generalisations*, and can be vague leaving room for discretion.

The study (5) defined complexity in terms of 'processing dificulty' in comprehending a regulation, such as its length, lexical diversity, use of conditional statements, and the overall readability of its sentences. A conclusion of the study notes the challenge of using machines to read regulatory text to increase efficiency and reduce cost as some texts are specifically vague and context-specific - hence the reliability of a machine learning model to correctly interpret regulatory text is of the utmost importance and needs to be proven.

A system that provides automated answers to questions on regulatory documents must ensure the answers are accurate. It is of fundamental importance that the answer is wholly answered based on a specific corpus of regulatory documents, and not influenced by other data sources. - this is of fundamental importance as regulatory question and answer system to provide accurate answers to questions is of fundamental importance. The phenomenon of AI hallucinations, where answers that are generated are factually incorrect, or even entirely

fabricated, must be avoided (6). A regulatory system that does not address the possibility of AI hallucinations is not reliable.

Whilst various technical mitigation strategies may be deployed to avoid hallucinations (such as improving training data quality, fine-tuning strategies, real-time fact-checking systems, hallucination detection algorithms (such as pattern recognition and confidence scoring), providing a citation on the document/section that is the source of the generated answer ensures a user can verify the answer based on the source text, which is considered the most reliable mitigation strategy, as it ensures the answer is validated by a human. Providing the citation for document/section/page, with potentially a link to open that precise source text, is reliable and saves significant time.

2.2 Sources and characteristics of regulatory documents

There are technical issue in sourcing the vast quantity of regulatory information that exists in different formats, locations, and in different languages (7). Documents are often not stored in computational formats (Resource Description Format (RDF), XML) and instead are in human-readable formats (typically PDF) requiring computerised interpretation. Document metadata is of vital importance (such as 'title', 'effective date' and 'version') for compiling a unique reference of documents, but this can be made complex when not available or not in standardised formats.

Legal texts can be very technical, and terminology in an EU Law context, can differ to that used at a national level (7).

2.3 Automated acquisition and preprocessing of regulatory documents

The Publications Office of the European Union hosts 'EUR-Lex' which is an online gateway to EU Law. It is the official source of EU legal documents, and includes the corpus of Regulations, Directives and Technical Standards that make up the European Banking Authority (EBA) Single Rulebook of regulatory standards applicable to all EU Banks (8).

The Interactive Single Rulebook (9) is an online compendium of regulatory documentation (level 1 text) as well as associated delegated and implementing acts, technical standards, guidelines and recommendations, with questions and answers.

The EUR-Lex online site enables a user to search for specific documents using a variety of search criteria , and search results include document metadata and a link to open the document itself in one of 24 languages and in a variety of formats. Additionally a web service is available enabling an external user, upon successful registration, to query the service directly, without having to use the website manually(10). Additional details are available on data extraction using web services (11). For the purpose of Exploratory Data

Analysis for this interim report the data was gathered manually, but automatic search and retrieval will be explored in the next stage of the project.

The benefit of using EUR-Lex over other document sources is that it is official, and has rich metadata describing each document, that will assist the identification of each document (unique numbering, name) as well as the relevance of each document (effective date, publication date, whether it is still in force).

Assuming requests are within limits, it should be possible to perform a background web service search of documents, compare to the list of documents already processed to identify any that are new, have been updated, or are no longer in force. An incremental download and processing of those documents will enable the regulatory document corpus used by the system to be refreshed. There are options in terms of the frequency. This could be a daily task, or may be sufficient once per week. The longer the frequency, the more data is likely to need to be processed which could limit performance. Testing during implementation will be able to determine optimum frequency.

It is also possible to retrieve document metadata and text in machine readable format (XML, RDF) using Cellar and Data Dump. These will also be explored as part of the project implementation.(12)

Other online sites such as Central Bank of Ireland, Financial Conduct Authority (UK), European Banking Authority have access to regulatory document information, but the EUR-Lex service is the authority for EU banking regulation, and hence the project will be limited in sourcing data from this reliable and official location.

2.4 Existing commercial solutions that support regulatory compliance

An extensive search of commercial solutions that support regulatory compliance was performed. It is clear that there is a rapidly expanding market for regulatory compliance platforms. These vary considerably in terms of functionality offered, and a detailed comparison has not been performed.

A key question is whether the system being developed as part of this project is already in widespread use in commercial applications. In general, it is believed that there is a key gap between existing systems and this project's proposed solution.

Where similar automated question and answer solutions exist based on a discrete set of regulatory documents, there remains concerns around accuracy, explainability and auditability.

On the basis of this review, this project's proposed system could be considered as augmented capability to existing solutions.

The following systems were identified and only briefly reviewed as part of the research.

Application: Facctum Overview: A centralised, automation-driven and AI-powered watchlist hub that ingests official regulatory sources, commercial providers, and internal high-risk records into one governed environment. Dashboards show which lists are in use, where, what has changed, and what is in development and what is in production.

Application: Resolver Overview: Automate repetitive workflows, track global regulatory changes in real time, and reduce compliance testing time by three quarters — all from one centralized compliance management platform. Get notified only when new requirements apply to your region, business type, or obligations. Instantly understand what changed and what action is needed, without combing through legal language or PDFs. Quickly spot similar or overlapping requirements, so you spend less time on repetitive work and keep controls consistent. When rules change, Resolver connects the compliance dots. You get full traceability from regulation to response — without chasing updates across teams. URL: https://www.resolver.com/grc-software/compliance-management

Application: Moody's Overview: A number of products that enable customer due diligence and identity verification during customer onboarding.

URL: https://www.moodys.com/web/en/us/kyc/resources/faq/risk-compliance.html

Application: Vcomply Overview: a centralized platform that enables banks to manage governance, risk, and compliance activities in an integrated and transparent way. It connects policies, controls, and regulatory obligations across departments, improving efficiency and accountability. By unifying these processes, banks can monitor risks in real time, adapt to regulatory changes, and maintain strong compliance oversight. URL: https://www.v-comply.com/blog/governance-risk-compliance-software-banks/

Application: LogicGate Risk Cloud Overview: A flexible, no-code GRC platform that enables banks to design workflows tailored to their risk and compliance needs. Its automation reduces manual tracking and accelerates decision-making. Foster scalable growth and ensure your organization operates within legal and regulatory boundaries while meeting internal and industry standards. Identify and address regulatory compliance gaps faster by linking obligations, assessments, and exams in one platform while automating workflows. URL: https://www.logicgate.com/

Application: SAI360 Overview: Comprehensive risk and compliance management with advanced analytics and regulatory framework support. Its modules cover operational risk, internal controls, and regulatory compliance. Banks and financial institutions face increasing regulatory pressure, operational complexity, and scrutiny from global stakeholders. SAI360 helps manage non-financial risk, ensure compliance with standards like Basel III and FFIEC, and strengthen internal controls—within a single, scalable bank GRC software platform. Track regulatory changes and assess impact across controls, policies, and operations. URL: https://www.sai360.com/

Application: Riskonnect Overview: Enterprise-level risk management that consolidates data

across departments, providing banks with actionable insights to control exposure and support strategic decisions. URL: https://riskonnect.com/en-gb/resources/risk-under-one-roof

Application: Aryza Overview: As regulatory demands increase, Aryza Unite offers a flexible and scalable financial compliance software solution to help you meet obligations with less manual effort and more insight. Automate breach reporting and action tracking, centralise compliance obligations, policies, and risk data in one place - connect them to relevant risks and controls. Built-in analytics and capital charge comparisons.

URL: https://www.aryza.com/

Application: Corporater Overview: Proactively Identify and Manage Enterprise Risks. Fully integrated and flexible software solution for organizations to assess risks and monitor the effectiveness of risk reduction activities across the enterprise.

URL: https://corporater.com/en/enterprise-risk-management-software

Application: SAP GRC Overview: SAP Governance, Risk, and Compliance (GRC) Solutions help organisations manage regulatory requirements, mitigate risks, and enforce internal policies. Designed for large enterprises, SAP GRC integrates seamlessly with existing SAP systems, providing a centralised platform for compliance and risk management. Key featuresinclude: Continuously evaluate and monitor risks across the organisation; Stay abreast of changing regulations and ensure adherence; Streamline internal audits with integrated tools. URL: https://www.sap.com/

Application: Archer Regulatory and Corporate Compliance Management Overview: Archer provides a centralised compliance platform designed to help organisations stay ahead of regulatory changes and manage risk effectively. With its customisable workflows and automated compliance tracking, Archer streamlines governance and corporate compliance processes. Key features include: Consolidate all compliance activities into a unified system; Generate reports effortlessly to demonstrate compliance; Develop, distribute, and track policies effectively. URL: https://www.archerirm.com

Application: Hyperproof Overview: Hyperproof is a compliance and risk management platform designed to simplify compliance workflows and ensure continuous monitoring. It helps organisations manage multiple frameworks in one place, reducing manual work and increasing efficiency. Key features include: Continuous compliance monitoring - Stay audit-ready at all times; Collaboration tools - Streamline teamwork with automated task management; Integration with existing systems - Connect with your tech stack for seamless compliance tracking. URL: https://hyperproof.io

Application: Diligent Overview: Diligent One Platformoffers an integrated platform for governance, risk, and compliance (GRC). It helps organisations automate audits, track risks, and ensure regulatory compliance. It's best for enterprises needing a scalable GRC solution with automation capabilities. Key features include: Identify and address potential risks

proactively; Simplify audits with automation and real-time data; Monitor compliance status and receive timely updates. URL: https://www.diligent.com/platform/diligent-one

Application: Thomson Reuters Compliance Management Overview: Thomson Reuters provides a comprehensive compliance platform that helps businesses stay compliant with evolving regulations through automation and real-time updates. This compliance management system is best for firms that require up-to-date regulatory insights and a trusted compliance platform. Key features include: Get real-time updates on compliance changes; Streamline compliance processes with automated tools; Track policies and ensure adherence. URL: https://legal.thomsonreuters.com/en/compliance

Application: Workiva Overview: Workiva is a cloud-based platform that streamlines complex reporting and compliance processes for organisations across various industries. It offers a unified solution for managing data, documents, and teams, ensuring accuracy and efficiency in compliance reporting. Key features include: Consolidate data from multiple sources to create comprehensive compliance reports; Enable teams to work simultaneously on documents, enhancing productivity and reducing errors; Link data across reports and presentations to ensure consistency and reduce manual updates. URL: https://www.workiva.com

Application: Complinity Overview: Complinity is a compliance software solution designed to help businesses and organizations manage their compliance obligations more efficiently and effectively. The software provides a centralized platform for tracking and monitoring compliance requirements, automating compliance workflows, and generating reports and analytics to help organizations make informed decisions. URL: https://complinity.com/

Application: Risk Hawk Overview: Risk Hawk is a risk management software that helps organizations identify, assess, and manage risks to achieve compliance with regulatory requirements. Risk Hawk aids organizations in staying up-to-date with regulatory changes and handling risks in ERM, IFC, ORM, and ICFR. It is available both on-premises and on cloud. It features a customizable regulatory compliance tracking software system, automated workflow and dashboards, and the ability to track compliance metrics with a single sign-on option. There is no free trial with RiskHawk. URL: https://www.riskhawk.net/

Application: Libryo Overview: By providing a centralized platform for risk management, Libryo has clients all over the world. With Libryo your legal compliance is ideal and is facilitated by automated workflows, robust analytics, and reporting. Libryo helps organizations improve the efficiency and effectiveness of their quality control and risk mitigation processes. URL: https://libryo.com/

Application: Predict 360 Overview: Predict360 is a powerful tool for organizations that need to manage complex risk and compliance obligations across multiple jurisdictions and industries. By providing a centralized platform for managing regulatory compliance, enterprise risk management, audit management, and regulatory change management processes, the software can help organizations reduce the risk of compliance issues and associated fines and

penalties, while also improving the efficiency and effectiveness of their risk and compliance management processes. URL: https://www.360factors.com/predict-360/

Application: NICE Actimize Overview: Financial crime, AML, surveillance, regulatory compliance for banks. Very widely used globally. URL: https://www.niceactimize.com/

Application: Wolters Kluwer OneSumX / Compliance Solutions Overview: Regulatory reporting, capital management, risk, compliance intelligence for financial institutions. Extremely relevant to EU banking regulation. Major presence in Ireland.

URL: https://www.wolterskluwer.com/en/solutions/onesumx-for-finance-risk-and-regulatory-reporting

Application: Ascent RegTech Overview: AI-driven regulatory change management using NLP to interpret regulations and map obligations. URL: https://www.ascentregtech.com/

Application: Regology Overview: Industry-agnostic regulatory intelligence platform quickly identifies all applicable regulatory content, monitors incoming changes, and generates obligations, risks, controls, and policies. The Regology platform enables compliance teams to leverage AI Agents to automate manual tasks, anticipate changes in the regulatory landscape, and act effectively to ensure compliance. URL: https://www.regology.com/

Application: CUBE Overview: CUBE monitors global regulatory activity in real-time and automatically updates policies and controls when relevant changes occur that impact your business. CUBE's AI engines and matching algorithms automatically identify common obligations across different regulators and jurisdictions.

URL: https://cube.global/sectors/requirement/obligations-management

Application: Corlytics Overview: Corlytics helps companies keep track of rules and regulations from around the world in one easy to use digital platform. With Corlytics you can confidently Find, Understand, Implement and Evidence regulatory compliance to the highest standards expected by regulators. Includes specific AI capabilities: Classification - AI driven categorisation of regulatory documents against a comprehensive taxonomy, rapidly identifying key themes and risks; Summarisation - delivering concise and accurate summaries of large, complex documents (regulations, regulatory alerts, policies etc.) Rationalisation & Mapping - identifying similarities across datasets to detect possible duplication and using identifiers and interconnected datasets for mapping and gap analysis. URL: https://www.corlytics.com/

Application: Fenergo Overview: Client lifecycle management, KYC, regulatory onboarding compliance. Widely used by banks. Irish company. URL: https://www.fenergo.com/

Application: ComplyAdvantage Overview: AI-driven AML risk intelligence, sanctions screening, regulatory monitoring. URL: https://kyc360.com/

Application: LexisNexis Regulatory Compliance / Lexis+ AI Overview: Legal research, regulatory intelligence, case law interpretation. URL: https://www.lexisnexis.co.uk/

Application: Westlaw (Thomson Reuters) Overview: Authoritative legal research database with AI capabilities.

URL: https://legalsolutions.thomsonreuters.co.uk/en/products-services/westlaw-uk.html

Application: AxiomSL Overview: Regulatory reporting software used by financial institutions. URL: https://www.nasdaq.com/solutions/fintech/nasdaq-axiomsl

2.5 AI approaches suitable for regulatory information retrieval

As part of the research for this project, it was necessary to develop an understanding of machine learning principles for unstructured data, and an understanding of how large language models can be used. A number of books have been invaluable in building a foundational knowledge that can be applied in this project (13) (14) (15) (16) (17) (18) (19).

A Large Language Model is trained on a very large source of text to be able to *understand* a language vocabulary and grammar. Text is broken down into small chunks called tokens, which are then converted into a numerical representations (vectors) that enable a large number of properties about each token to be defined, such as its similarities to other tokens through pattern matching (for example clustering). These vector embeddings are therefore meaningful when associated with their large language model, which holds all embedding vectors for all tokens.

The introduction of Bidirectional Encoder Representations from Transformers (BERT) in 2018 enabled *semantic search* as opposed to simply *keyword search* capability, which enabled search based on the meaning of text for the first time. When combined with generative large language models this gave the ability to ask questions and receive answers based on the semantic search request. An issue was that the generated answers were not always correct, known as "Hallucinations". The work around was to have the model's answers generated from just relevant source information, known as Retrieval-Augmented Generation (RAG).

3 Methods and Exploration of Data

3.1 Ethical consideration of source data

The data used in the project is publicly available regulatory text from the European Union. There is no ethical constraint to use this data. There is, however, an important ethical consideration in the way the data will be used to automatically answer questions on regulations. It is vitally important that there is traceability of the source of an automated response, enabling the user to validate the response that has been given directly by inspecting the text.

This is a way of mitigating any error in the response generated and eliminating potential consequences of making decisions on the answer alone, without validating its accuracy.

3.2 Conceptual Design

A conceptual end-to-end flow has been designed (Refer Figure 1).

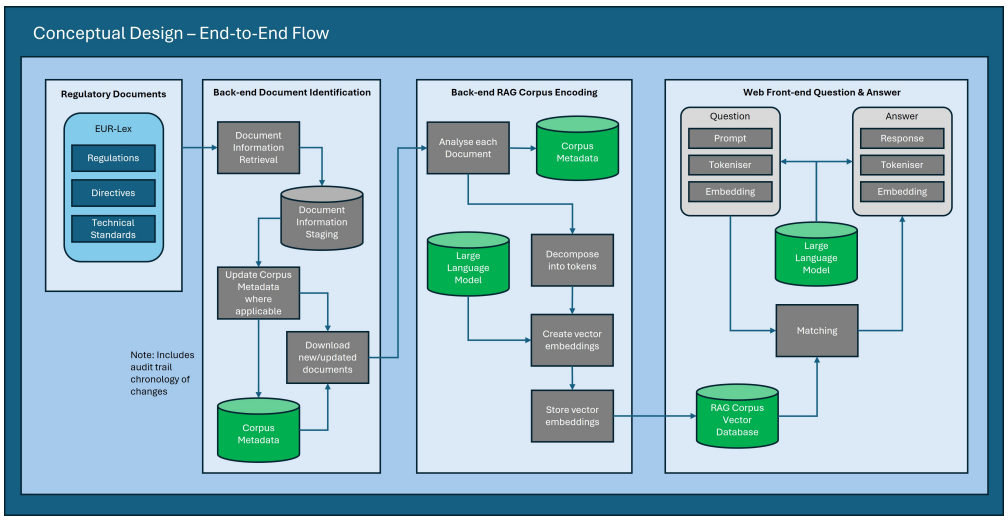

Figure 1: Conceptual Design - End-to-End Flow

3.3 Preliminary Exploratory Data Analysis

The EUR-Lex site was used to search for banking regulation using the following search criteria

- Regulation = TRUE
- Corrigenda = TRUE
- Consolidated Versions = FALSE
- Theme = (2416) financial institutions & credit

The complete metadata for the first 80 documents was exported in csv format "ExampleEUR-LexSearchResults.csv".

Ten random documents were downloaded from EUR-Lex to perform preliminary Exploratory Data Analysis (refer separate python notebook).

The filenames were loaded to a dataframe based on a file list from a google drive location. Document text was extracted and analysed for word count and page count using pdfplumber. No errors were detected.

Basic analysis was performed to determine variability between documents in terms of word count and document length.

Document text was converted to lower case, and special characters were removed, as well as stopwords > 3 characters in length were removed. Word frequency per document was analysed. Cosine similarity was performed to show the similarity of words used per document and a heatmap charted the results (Refer Figure 2).

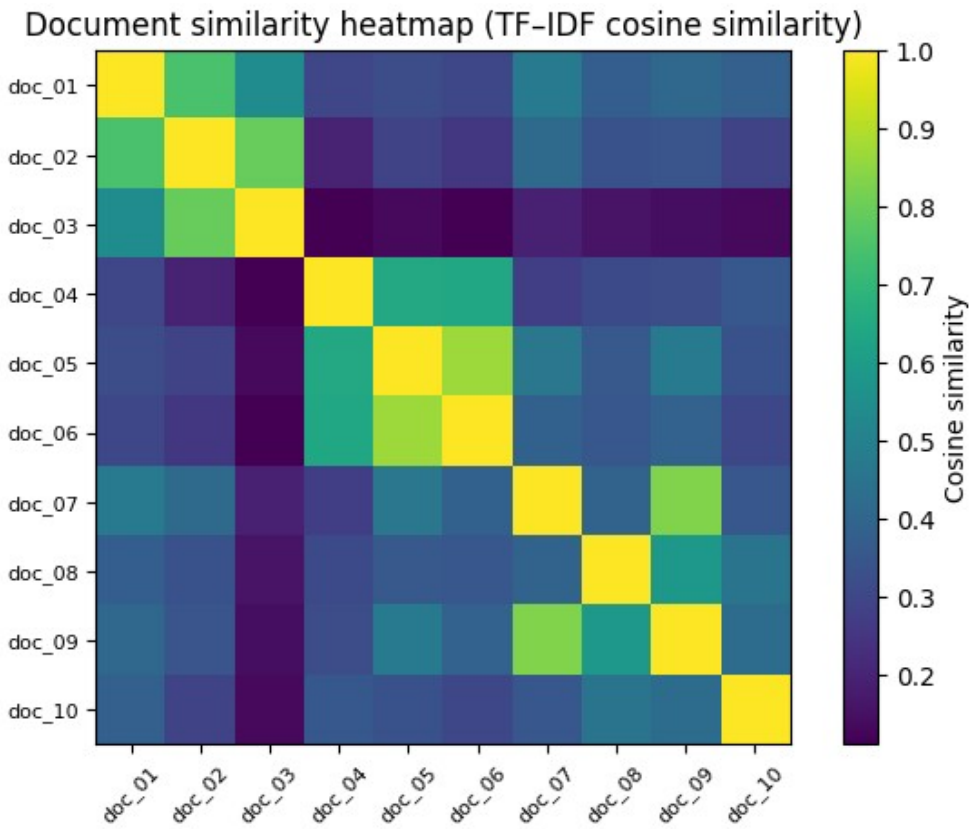

Figure 2: Cosine Similarity of Document Words

Principal Component Analysis was then used to reduce attributes to two dimensions enabling enabling visualisation of clusters (Refer Figure 3).

4 Proposed Future Analysis

The remainder of the project will build on the work performed to date to develop a working prototype application.

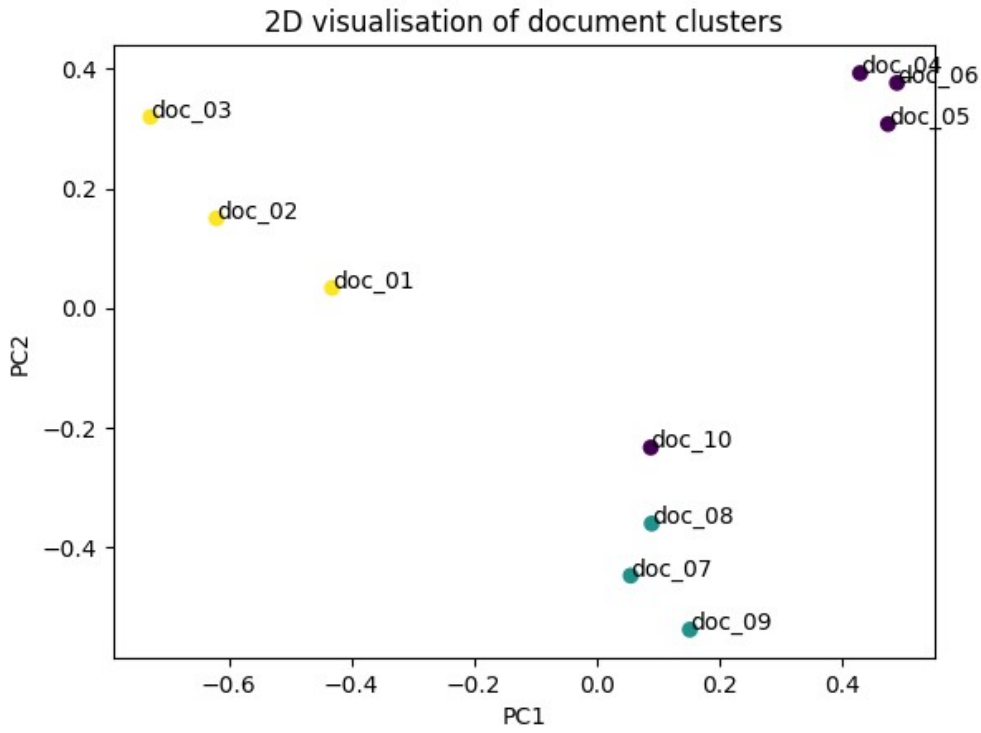

Figure 3: Visualisation of Clusters Based on Document Words

4.1 Project Plan

The project duration is 15 weeks in total.

The Project Brief was agreed 29th January. Since then, four weeks have elapsed, and preliminary work covered to date, up to this interim report, is as follows:

Week01 2-February:

- Develop understanding of Large Language Models and Machine Learning for Unstructured Data through seven books (13) (14) (15) (16) (17) (18) (19).

Week02 9-February:

- Continue reading books on Large Language Models and Unstructured Data
- Develop initial conceptual design and end-to-end flow

Week03 16-February:

- Less progress during the week than desired due to high activity at work
- Initial meeting with project supervisor

- Research relevant commercial regulatory compliance software applications
- Research how to source official regulatory documents and available document metadata

Week04 23-February:

- Less progress during the week than desired due to high activity at work
- Unable to meet Project Supervisor for the second meeting due to issue at work
- Refine conceptual design and end-to-end flow
- Exploratory data analysis of ten randomly selected regulatory documents
- Files loaded to GitHub (https://github.com/GeorgeSemain/PROJC9013).
- Submission of Interim Report

There are 11 weeks to complete the project, and the activities have been sequenced per week as follows:

Week05 2-March:

- Select 100 documents from EUR-Lex and perform EDA

Week06 9-March:

- Sign-up to EUR-Lex and test web service direct access to document metadata

Week07 16-March:

- Test automatic retrieval of documents from EUR-Lex based on metadata
- Update section in final report

Week08 23-March:

- Develop RAG embeddings of 100 documents

Week09 30-March:

- Develop initial 'Question and Answer' approach in Python notebook

Week10 6-April:

- Test initial 'Question and Answer' approach in Python notebook and explore alternative models
- Hypothesis testing
- Determine how RAG embeddings will be stored and accessed by users
- Update section in final report

Week11 13-April:

- Test including document citation as part of response
- Explore potential MongoDB storage of document metadata and consider others
- Determine how to store and access document metadata

Week12 20-April:

- Test ability to identify conflicting text and including in the response
- Detailed design of application modules for coding in Shiny for Python

Week13 27-April:

- Application build in Shiny for Python

Week14 4-May:

- Application build and test in Shiny for Python
- Update final report - complete draft
- Review report with project supervisor

Week15 11-May:

- Update final report and submit

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